

Chapter 3

Course outline Bivariate Distributions, Marginal Distributions, Conditional Distributions and independence, Conditional expectation and conditional variance. Bivariate normal distribution and its properties

1. PROBABILITY DENSITY FUNCTION

The function $f(x)$ is called a probability density function for a continuous random variable X if

- a) If the total area under the curve is equal to one.
- b) The area under the curve between any two points $x=a$ and $x=b$ gives the probability that X lies between a and b .

PROPERTIES OF PROBABILITY DENSITY FUNCTION

1. $f(x) \geq 0$ for all x
2. $\int_{-\infty}^{\infty} f(x) dx = 1$
3. $P(a < X < b) = \int_a^b f(x) dx$

2. DISTRIBUTION FUNCTION

The distribution function for a continuous random variable X is defined as

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx \quad -\infty < x < \infty$$

If $F(x)$ is continuous at the point x and its first derivative exists

$$\frac{dF(x)}{dx} = f(x)$$

PROPERTIES OF DISTRIBUTION FUNCTION

- i) If $y > x$, then $F(y) \geq F(x)$, $F(x)$ is a non-decreasing function.
- ii) $F(-\infty) = 0$ and $F(\infty) = 1$

$$\text{iii)} \quad P(X \geq x) = \int_x^{\infty} f(x)dx = 1-F(x)$$

$$\text{iv)} \quad P(a < X < b) = \int_a^b f(x)dx = F(b) - F(a)$$

Note: the distribution function is also called cumulative distribution function (c. d. f)

3. Bivariate Continuous Distributions

Let X and Y be two random variables then $P(X \leq x, Y \leq y) = F(x, y)$ is defined as the joint distribution function or cumulative distribution function (c. d. f) of these random variables, we have

$$F(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, u) dt du$$

The distribution function of X is

$$P(X \leq x) = F_1(x) = F(x, \infty) = \int_{-\infty}^{\infty} \int_{-\infty}^x f(x, y) dx dy$$

The distribution function of Y is

$$P(Y \leq y) = F_2(y) = F(\infty, y) = \int_{-\infty}^{\infty} \int_{-\infty}^y f(x, y) dx dy$$

PROPERTIES OF BIVARIATE DISTRIBUTION FUNCTION

$$\text{i)} \quad 0 \leq F(x, y) \leq 1$$

$$\text{ii)} \quad F(x, -\infty) = F(-\infty, y) = 0, \quad F(\infty, \infty) = 1$$

$$\text{iii)} \quad \text{If } x_1 < x_2 \text{ then}$$

$$P(X \leq x_2, Y \leq y) = P(X \leq x_1; Y \leq y) + P(x_1 < X \leq x_2; Y \leq y)$$

And hence

$$P(x_1 < X \leq x_2; Y \leq y) = F(x_2, y) - F(x_1, y) \geq 0$$

Similarly if

$$\text{If } y_1 < y_2 \text{ then}$$

$$P(X \leq x_2, y_1 < Y \leq y_2) = F(x, y_2) - F(x, y_1) \geq 0$$

It follows $F(x, y)$ is non decreasing function of x and y

4. JOINT DENSITY FUNCTION

The bivariate probability density function of continuous random variable X and Y is an integrable function $f(x, y)$ satisfying the following properties

- i) $f(x, y) \geq 0$ for all (x, y)
- ii) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$
- iii) $P(a \leq X \leq b, c \leq Y \leq d) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy$
- iv) $\frac{\partial F(x, y)}{\partial x \partial y} = f(x, y)$ where ever F is differentiable

5. MARGINAL PROBABILITY DENSITY FUNCTION

The marginal p. d. f of the continuous random variable X is

$$f_1(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

The marginal p. d. f of the continuous random variable Y is

$$f_2(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

That is the marginal p. d. f of any random variables is obtained by integrating out the other variable from the joint p. d. f between the limits $-\infty$ and ∞

6. CONDITIONAL PROBABILITY DENSITY FUNCTION

The conditional p. d. f of the continuous random variable X given that Y takes the value y is defined as

$$f(x|y) = \frac{f(x, y)}{f_2(y)}$$

Where $f(x, y)$ = joint probability density function of X and Y

$f_1(y)$ = marginal probability density function of Y

Similarly the conditional p. d. f of the continuous random variable Y given that X takes the value x is defined as

$$f(y|x) = \frac{f(x,y)}{f_2(y)}$$

Where $f(x, y)$ = joint probability density function of X and Y

$f_1(x)$ = marginal probability density function of X

7. STATISTICAL INDEPENDENCE

Two random variables X and Y are said to be independent if

$$f(x, y) = f_1(x)f_2(y)$$

The two random variable X and Y with c. d. f $F(x, y)$ are said to be independent if $F(x, y) = F_1(x)F_2(y)$

Or If X and Y are independent then

$$f(x/Y=y) = f_1(x) \quad \text{and} \quad f(y/X=x) = f_2(y)$$

8. Expectation and Moments

The expectation of a function $g(x, y)$ is defined as

$$E[g(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy$$

provided the integral converges

The bivariate moments of the joints distribution about the origin are defined as

$$\mu'_{r,s} = E[X^r Y^s] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^r y^s f(x, y) dx dy$$

Where r, s are integers and the sum $r + s$ indicates the order of the moments.

In particular $\mu'_{r,0} = E(X^r)$ $\mu'_{0,s} = E(Y^s)$

$$\mu'_{1,0} = \bar{x} \quad \mu'_{0,1} = \bar{y}$$

The bivariate mean moments about mean are defined as

$$\mu_{rs} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \bar{x})^r (y - \bar{y})^s f(x, y) dx dy$$

$$\sigma_x^2 = \mu_{20} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x, y) dx dy$$

$$\sigma_y^2 = \mu_{02} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (y - \bar{y})^2 f(x, y) dx dy$$

The Covariance of X and Y is defined as

$$\begin{aligned} \text{Cov.}(X, Y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \bar{x})(y - \bar{y}) f(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y) dx dy - \bar{x} \bar{y} \end{aligned}$$

If X and Y are independent then

$$\mu'_{r,s} = \mu'_{r,0} \times \mu'_{0,s}$$

$$\mu_{r,s} = \mu_{r,0} \times \mu_{0,s}$$

Bivariate Moment Generating Function

The moments generating function $M_0(t_1, t_2)$ of a bivariate continuous distribution about the origin is defined as

$$M_0(t_1, t_2) = E(e^{t_1 X + t_2 Y}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{t_1 x + t_2 y} f(x, y) dx dy$$

Rth order moments about origin = Coefficient of $\frac{t_1^r t_2^s}{r! s!}$

Similarly the mean moments generating function is defined as

$$M_{r,s}(t_1, t_2) = e^{-t_1 m_1 - t_2 m_2} M_0(t_1, t_2)$$

Conditional Expectation

If $f(x, y)$ is the joint density function of two continuous random variable and the conditional density of Y given $X = x$ is $f(y/X=x)$ and $f_1(x)$ is marginal density of X then Conditional expectation is defined as

$$E(Y/X) = \int_{-\infty}^{\infty} y f(y|x) dy = \int_{-\infty}^{\infty} y \frac{f(x,y)}{f_1(x)} dy = m_2(x)$$

$$= \beta_0 + \beta_2 x$$

And is a function of x and is called the regression function of y on x.

Similarly

$$E(X/Y) = \int_{-\infty}^{\infty} x f(x|y) dx = \int_{-\infty}^{\infty} x \frac{f(x,y)}{f_2(y)} dx = m_1(y)$$

$$= \beta_0 + \beta_1 y$$

(The regression function of x on y).

S The correlation coefficient is

$$r = \sqrt{\beta_1 \beta_2}$$

Conditional Variance

If $f(x, y)$ is the joint density function of two continuous random variable and the conditional density of Y given $X = x$ is $f(y/X=x)$ and $f_1(x)$ is marginal density of X then Conditional Variance is defined as

$$\text{Var}(Y|X=x) = \int_{-\infty}^{\infty} [y - m_2(x)]^2 \frac{f(x,y)}{f_1(x)} dy$$

(sometimes called residual variance of Y on X)

Similarly

$$\text{Var}(X|Y=y) = \int_{-\infty}^{\infty} [x - m_1(y)]^2 \frac{f(x,y)}{f_2(y)} dx$$

Important Results

$$\text{i) } E[E(Y|X)] = E(Y) \quad E[E(X|Y)] = E(X)$$

$$\text{ii) } \text{Var } X = E[\text{Var}(X|Y)] + \text{Var}[E(X|Y)]$$

Problems on Bivariate Distributions

Q#1. Given the following joint p. d. f

$$f(x, y) = \frac{1}{8}(6 - x - y) \quad 0 \leq x \leq 2, 2 \leq y \leq 4$$

$$= 0 \quad \text{elsewhere}$$

- Verify that $f(x, y)$ is a joint density function.
- (i) Calculate $P(X \leq \frac{3}{2}, Y \leq \frac{5}{2})$, (ii) $P(X+Y < 3)$
- Find the marginal p. d. f of X and Y
- Find the conditional p. d. f $f(x|y)$ and $f(y|x)$

Solution

a) $f(x, y)$ will be p. d. f if (i) $f(x, y) > 0$ for all X and Y

$$(ii) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

First Condition is obvious and second condition is

$$\begin{aligned} \text{Total area} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy \\ &= \frac{1}{8} \int_0^2 \int_2^4 (6 - x - y) dx dy \\ &= \frac{1}{8} \int_0^2 \left[6y - xy - \frac{y^2}{2} \right]_2^4 dx \\ &= \frac{1}{8} \int_0^2 [\{6 \times 4 - 4x - 8\} - \{6 \times 2 - 2x - 2\}] dx \\ &= \frac{1}{8} \int_0^2 [6 - 2x] dx \\ &= \frac{1}{8} \left[6x - 2 \frac{x^2}{2} \right]_0^2 \\ &= \frac{1}{8} (6 \times 2 - 2^2) \\ &= 1 \end{aligned}$$

Hence $f(x, y)$ is a proper p. d. f

$$\begin{aligned}
 \text{b) } P(X \leq \frac{3}{2}, Y \leq \frac{5}{2}) &= \frac{1}{8} \int_0^{3/2} \int_2^{5/2} (6 - x - y) dx dy \\
 &= \frac{1}{8} \int_0^{3/2} \left[6y - xy - \frac{y^2}{2} \right]_2^{5/2} dx \\
 &= \frac{1}{8} \int_0^{3/2} \left[\left\{ 15 - \frac{5}{2}x - \frac{25}{8} \right\} - \{ 12 - 2x - 2 \} \right] dx \\
 &= \frac{1}{8} \int_0^{3/2} \left[\frac{15}{8} - \frac{1}{2}x \right] dx \\
 &= \frac{1}{8} \left[\frac{15}{8}x - \frac{1}{4}x^2 \right]_0^{3/2} \\
 &= \frac{1}{8} \left[\frac{45}{16} - \frac{9}{16} \right] = \frac{9}{32} = 0.28
 \end{aligned}$$

$$\begin{aligned}
 P(X+Y < 3) &= \frac{1}{8} \int_0^1 \int_2^{3-x} (6 - x - y) dx dy \\
 &= \frac{1}{8} \int_0^1 \left[6y - xy - \frac{y^2}{2} \right]_2^{3-x} dx \\
 &= \frac{1}{8} \int_0^1 \left[\left\{ 6(3-x) - x(3-x) - \frac{(3-x)^2}{2} \right\} - \{ 6(2) - 2x - 2 \} \right] dx \\
 &= \frac{1}{8} \int_0^1 \left[\left\{ 18 - 6x - 3x + x^2 - \frac{9+x^2-6x}{2} \right\} - \{ 10 - 2x \} \right] dx \\
 &= \frac{1}{8} \int_0^1 \left(18 - 9x + x^2 - \frac{9+x^2-6x}{2} - 10 + 2x \right) dx \\
 &= \frac{1}{8} \int_0^1 \left(8 - 7x + x^2 - \frac{9+x^2-6x}{2} \right) dx \\
 &= \frac{1}{16} \int_0^1 (16 - 14x + 2x^2 - 9 - x^2 + 6x) dx \\
 &= \frac{1}{16} \int_0^1 (x^2 - 8x + 7) dx \\
 &= \frac{1}{16} \left[\frac{x^3}{3} - 8\frac{x^2}{2} + 7x \right]_0^1 \\
 &= \frac{1}{16} \left[\frac{1}{3} - 4 + 7 \right]_0^1 \\
 &= \frac{1}{16} \times \frac{10}{3} = \frac{10}{48}
 \end{aligned}$$

$$P(X+Y < 3) = \frac{5}{24}$$

c) Marginal p. d. f of X

$$\begin{aligned}
 f_1(x) &= \frac{1}{8} \int_2^4 (6 - x - y) dy & 0 < x < 2 \\
 &= \frac{1}{8} \left[6y - xy - \frac{y^2}{2} \right]_2^4 \\
 &= \frac{1}{8} [\{24 - 4x - 8\} - \{12 - 2x - 2\}] \\
 &= \frac{1}{8} [6 - 2x] & 0 < x < 2 \\
 &= \frac{1}{4} [3 - x] & 0 < x < 2 \\
 &= 0 & \text{elsewhere}
 \end{aligned}$$

Marginal p. d. f of Y

$$\begin{aligned}
 f_2(y) &= \frac{1}{8} \int_0^2 (6 - x - y) dx & 2 < y < 4 \\
 &= \frac{1}{8} \left[6x - \frac{x^2}{2} - xy \right]_0^2 \\
 &= \frac{1}{8} [\{12 - 2 - 2y\}] \\
 &= \frac{1}{8} [10 - 2x] & 2 < y < 2 \\
 &= \frac{1}{4} [5 - x] & 2 < y < 2 \\
 &= 0 & \text{elsewhere}
 \end{aligned}$$

d) Find the conditional p. d. f $f(x|y)$ and $f(y|x)$ The conditional p. d. f $f(x|y)$

$$\begin{aligned}
 f(x|y) &= \frac{f(x,y)}{f_2(y)} \\
 &= \frac{\frac{1}{8}(6-x-y)}{\frac{1}{4}[5-x]} \\
 &= \frac{(6-x-y)}{2[5-x]}
 \end{aligned}$$

The conditional p. d. f $f(y|x)$

$$\begin{aligned}
 f(y|x) &= \frac{f(x,y)}{f_1(x)} \\
 &= \frac{\frac{1}{8}(6-x-y)}{\frac{1}{4}[3-x]} \\
 &= \frac{(6-x-y)}{2[3-x]}
 \end{aligned}$$

Q#2. Let the Bivariate Continuous random variables (X, Y) have the joint p.d.f given by

$$f(x, y) = x^2 + \frac{xy}{3}, \quad 0 \leq x \leq 1, 0 \leq y \leq 2$$

$$= 0 \quad \text{elsewhere}$$

- Check that $f(x, y)$ is a p. d. f
- Find the marginal p. d. f's
- Find the conditional p. d. f's and verify that $f(x | y)$ is a p. d. f

Solution

a) $f(x, y)$ will be p. d. f if (i) $f(x, y) > 0$ for all X and Y

$$(ii) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

First Condition is obvious and second condition is

$$\text{Total area} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy$$

$$= \int_0^1 \int_0^2 (x^2 + \frac{xy}{3}) dx dy$$

$$= \int_0^1 \left[x^2 y + \frac{xy^2}{6} \right]_0^2 dx$$

$$= \int_0^1 \left[2x^2 + \frac{2}{3} x \right] dx$$

$$= \left[2 \frac{x^3}{3} + \frac{2}{3} \frac{x^2}{2} \right]_0^1$$

$$= \frac{2}{3} + \frac{1}{3}$$

$$= 1$$

Hence $f(x, y)$ is a proper p. d. f

b) Marginal p. d. f of X is

$$f_1(x) = \int_0^2 (x^2 + \frac{xy}{3}) dy \quad 0 < x < 1$$

$$= \left[x^2 y + \frac{xy^2}{2} \right]_0^2$$

$$= 2x^2 + 2x$$

$$= 2x(x+1)$$

$$0 < x < 1$$

$$= 0$$

elsewhere

Marginal p. d. f of Y is

$$f_2(y) = \int_0^1 \left(x^2 + \frac{xy}{3} \right) dx$$

$$2 < y < 4$$

$$= \left[\frac{x^3}{3} + \frac{x^2 y}{6} \right]_0^1$$

$$= \frac{1}{3} + \frac{y}{6}$$

$$= \frac{1}{6}(2 + y)$$

$$2 < y < 4$$

$$= 0$$

elsewhere

The conditional p. d. f $f(x|y)$

$$f(x|y) = \frac{f(x,y)}{f_2(y)}$$

$$= \frac{x^2 + \frac{xy}{3}}{\frac{1}{6}(2+y)} = \frac{6x^2 + 2xy}{2+y}$$

$$0 < x < 1, \quad 2 < y < 4$$

Check $f(x|y)$ is a p. d. f

$$\text{Toto area} = \int_{-\infty}^{\infty} f(x|y) dx$$

$$= \int_0^1 \left(\frac{6x^2 + 2xy}{2+y} \right) dx$$

$$= \frac{1}{2+y} \left[6 \frac{x^3}{3} + 2 \frac{x^2 y}{2} \right]_0^1$$

$$= \frac{1}{2+y} [2 + y]$$

$$= 1. \quad \text{For all values of } y$$

Q#3. Suppose the joint p.d.f of (X,Y) is given by

$$f(x, y) = x^2 + \frac{xy}{3}, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 2$$

$$= 0 \quad \text{elsewhere}$$

a) Check that $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$

b) Compute (i) $P(X > \frac{1}{2})$ (ii) $P(Y < X)$ (iii) $P(X + Y > 1)$

(iv) $P(Y < \frac{1}{2} \mid X < \frac{1}{2})$

Solution

$$a) \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = \int_0^1 \int_0^2 \{x^2 + \frac{xy}{3}\} dx dy$$

$$= \int_0^1 \left[x^2 y + \frac{xy^2}{6} \right]_0^2 dx$$

$$= \int_0^1 \left[2x^2 + \frac{2x}{3} \right] dx$$

$$= \left[\frac{2x^3}{3} + \frac{2x^2}{6} \right]_0^1$$

$$= \frac{2}{3} + \frac{1}{3}$$

$$= 1$$

$$i) \quad P(X > \frac{1}{2}) = \int_{1/2}^1 \int_0^2 (x^2 + \frac{xy}{3}) dx dy$$

$$= \int_{1/2}^1 \left[x^2 y + \frac{xy^2}{6} \right]_0^2 dx$$

$$= \int_{1/2}^1 \left[2x^2 + \frac{2x}{3} \right] dx$$

$$= \left[\frac{2x^3}{3} + \frac{x^2}{3} \right]_{1/2}^1$$

$$= \left[\left\{ \frac{2}{3} + \frac{1}{3} \right\} - \left\{ \frac{1}{12} + \frac{1}{12} \right\} \right]$$

$$= 5/6$$

$$ii) \quad P(Y < X) = \int_0^1 \int_0^x (x^2 + \frac{xy}{3}) dx dy$$

$$= \int_0^1 \left[x^2 y + \frac{xy^2}{6} \right]_0^x dx$$

$$\begin{aligned}
&= \int_0^1 \left[x^3 + \frac{x^3}{6} \right] dx \\
&= \int_0^1 \left(\frac{7x^3}{6} \right) dx \\
&= \left[\frac{7x^4}{6(4)} \right]_0^1 \\
&= \frac{7}{24}
\end{aligned}$$

$$\begin{aligned}
\text{iii) } P(X+Y > 1) &= 1 - P(X+Y < 1) \\
&= 1 - \int_0^1 \int_0^{1-x} \left(x^2 + \frac{xy}{3} \right) dx dy \\
&= 1 - \int_0^1 \left[x^2 y + \frac{xy^2}{6} \right]_0^{1-x} dy \\
&= 1 - \int_0^1 \left[x^2 - x^3 + \frac{x+x^3-2x^2}{6} \right] dx \\
&= 1 - \frac{1}{6} \int_0^1 [4x^2 - 5x^3 + x] dx \\
&= 1 - \frac{1}{6} \left[4 \frac{x^3}{3} - 5 \frac{x^4}{4} + \frac{x^2}{2} \right]_0^1 \\
&= 1 - \frac{1}{6} \left(\frac{7}{12} \right) \\
&= 1 - \frac{7}{72} = \frac{65}{72}
\end{aligned}$$

$$\text{iv) } P\left(Y < \frac{1}{2} \mid X < \frac{1}{2}\right) = \frac{P(X < \frac{1}{2}, Y < \frac{1}{2})}{P(X < \frac{1}{2})} \quad \text{----- (i)}$$

$$\begin{aligned}
f_1(x) &= \int_0^2 \left(x^2 + \frac{xy}{3} \right) dy \\
&= \left[x^2 y + \frac{xy^2}{6} \right]_0^2 \\
&= 2x^2 + \frac{2x}{3}
\end{aligned}$$

$$\begin{aligned}
P(X < \frac{1}{2}) &= \int_0^{\frac{1}{2}} \left\{ 2x^2 + \frac{2x}{3} \right\} dx \\
&= \left[\frac{2x^3}{3} + \frac{2x^2}{6} \right]_0^{\frac{1}{2}} \\
&= \frac{2}{3} * \frac{1}{8} + \frac{2}{6} * \frac{1}{4} = 1/6
\end{aligned}$$

$$\begin{aligned}
P(X < 1/2, Y < 1/2) &= \int_0^{1/2} \int_0^{1/2} \left(x^2 + \frac{xy}{3} \right) dx dy \\
&= \int_0^{1/2} \left[\frac{x^3}{3} + \frac{xy^2}{6} \right]_0^{1/2} dy
\end{aligned}$$

$$= \int_0^{1/2} \left[\frac{1}{24} + \frac{x}{24} \right] dx = \frac{1}{24} \int_0^{1/2} (1+x) dx$$

$$= \frac{1}{24} \left[x + \frac{x^2}{2} \right]_0^{1/2} = \frac{1}{24} \left[\frac{1}{2} + \frac{1}{8} \right] = 5/192$$

$$P(Y < \frac{1}{2} \mid X < \frac{1}{2}) = \frac{7/192}{1/12} = 5/16$$

Q#4. If the two random variables X and Y have the joint p. d. f

$$f(x, y) = \frac{2}{3} (x + 2y) \quad 0 \leq x \leq 1, 0 \leq y \leq 1$$

$$= 0 \quad \text{elsewhere}$$

a) Find the marginal distribution of X and Y, and their conditional distribution. Find $P(X < \frac{1}{2} \mid Y = \frac{1}{2})$

Solution

Marginal distribution of X	Marginal distribution of Y
$f_1(x) = \int_{-\infty}^{\infty} f(x, y) dy$ $= \frac{2}{3} \int_0^1 (x + 2y) dy$ $= \frac{2}{3} \left[xy + \frac{2y^2}{2} \right]_0^1$ $= \frac{2}{3} (x + 1)$	$f_2(y) = \int_{-\infty}^{\infty} f(x, y) dx$ $= \frac{2}{3} \int_0^1 (x + 2y) dx$ $= \frac{2}{3} \left[\frac{x^2}{2} + 2xy \right]_0^1$ $= \frac{2}{3} \left[\frac{1}{2} + 2y \right]$ $= \frac{1}{3} (1 + 4y)$

Conditional p.d.f of Y given X	Conditional p. d. f of X given Y
$f(y x) = \frac{f(x, y)}{f_1(x)}$ $= \frac{\frac{2}{3}(x+2y)}{\frac{2}{3}(x+1)}$ $= \frac{x+2y}{x+1}$	$f(x y) = \frac{f(x, y)}{f_2(y)}$ $= \frac{\frac{2}{3}(x+2y)}{\frac{1}{3}(1+4y)}$ $= \frac{2(x+2y)}{(1+4y)}$

$$f(x|y) = \frac{2(x+2y)}{(1+4y)}$$

$$\begin{aligned} P\left(X < \frac{1}{2} \mid y = \frac{1}{2}\right) &= \int_0^{1/2} f\left(x \mid y = \frac{1}{2}\right) dx \\ &= \int_0^{1/2} \frac{2(x+2(\frac{1}{2}))}{1+4(\frac{1}{2})} dx = \int_0^{1/2} \frac{2}{3} (x+1) dx \\ &= \frac{2}{3} \left[\frac{x^2}{2} + x \right]_0^{1/2} = \frac{2}{3} \left[\frac{1}{8} + \frac{1}{2} \right] \\ &= \frac{5}{12} \end{aligned}$$

Q# 5. Given the joint p. d. f $f(x, y) = 3xy(x + y)$ $0 \leq x \leq 1, 0 \leq y \leq 1$
 $= 0$ elsewhere

Find the marginal p. d. f's and conditional p. d. f. Also find the conditional.

Probability $P\left[\frac{1}{2} \leq X \leq \frac{3}{4} \mid \frac{1}{2} < Y < \frac{2}{3}\right]$

Solution

Marginal p. d. f of X is

$$\begin{aligned} f_1(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\ &= 3 \int_0^1 xy(x + y) dy \\ &= 3 \left[x^2 \frac{y^2}{2} + \frac{xy^3}{3} \right]_0^1 = \frac{1}{2} (3x^2 + 2x) \end{aligned}$$

Marginal p. d. f of Y is

$$\begin{aligned} f_2(y) &= \int_{-\infty}^{\infty} f(x, y) dx \\ &= 3 \int_0^1 xy(x + y) dx \end{aligned}$$

$$= 3 \left[y^2 \frac{x^2}{2} + \frac{y x^3}{3} \right]_0^1 = \left(\frac{3}{2} y^2 + y \right)$$

Conditional p. d. f of Y given X

$$\begin{aligned} f(y|x) &= \frac{f(x,y)}{f_1(x)} \\ &= \frac{3xy(x+y)}{\frac{1}{2}(3x^2+2x)} = \frac{6xy(x+y)}{(3x^2+2)} = \frac{6y(x+y)}{(3x+2)} \end{aligned}$$

Conditional p. d. f of X given Y

$$\begin{aligned} f(y|x) &= \frac{f(x,y)}{f_2(y)} \\ &= \frac{3xy(x+y)}{\frac{1}{2}(3y^2+2y)} = \frac{6x(x+y)}{(3y+2)} \end{aligned}$$

$$P\left[\frac{1}{2} \leq X \leq \frac{3}{4} \mid \frac{1}{2} < Y < \frac{2}{3}\right] = \frac{P\left(\frac{1}{2} < X < \frac{3}{4}, \frac{1}{2} < Y < \frac{2}{3}\right)}{P\left(\frac{1}{2} < Y < \frac{2}{3}\right)} \quad (i)$$

$$\begin{aligned} P\left(\frac{1}{2} < Y < \frac{2}{3}\right) &= \int_{1/2}^{2/3} f_2(y) dy \\ &= \int_{1/2}^{2/3} \left(\frac{3}{2}y^2 + y\right) dy \\ &= \left[\frac{3y^3}{2 \cdot 3} + \frac{y^2}{2}\right]_{1/2}^{2/3} = \frac{1}{2} \left[\left\{\frac{8}{27} + \frac{4}{9}\right\} - \left\{\frac{1}{8} + \frac{1}{4}\right\}\right] = 79/432 \end{aligned}$$

$$\begin{aligned} P\left(\frac{1}{2} < X < \frac{3}{4}, \frac{1}{2} < Y < \frac{2}{3}\right) &= \int_{\frac{1}{2}}^{\frac{3}{4}} \int_{\frac{1}{2}}^{\frac{2}{3}} 3xy(x+y) dx dy \\ &= 3 \int_{1/2}^{3/4} \left[x^2 \frac{y^2}{2} + x \frac{y^3}{3} \right]_{1/2}^{2/3} dx \\ &= 3 \int_{1/2}^{3/4} \left[\left\{ \frac{4}{18} x^2 + \frac{8}{81} x \right\} - \left\{ \frac{1}{8} x^2 + \frac{1}{24} x \right\} \right] dx \\ &= 3 \int_{1/2}^{3/4} \left\{ \frac{7}{72} x^2 + \frac{37}{648} x \right\} dx \end{aligned}$$

$$\begin{aligned}
&= 3 \left[\frac{7}{72} \frac{x^3}{3} + \frac{37}{648} \frac{x^2}{2} \right]_{1/2}^{3/4} \\
&= 3 * \left[\left\{ \frac{7}{72} \frac{27}{192} + \frac{37}{648} \frac{9}{32} \right\} - \left\{ \frac{7}{72} \frac{1}{24} + \frac{37}{648} \frac{1}{8} \right\} \right] \\
&= 769/13824
\end{aligned}$$

$$P\left[\frac{1}{2} \leq X \leq \frac{3}{4} \mid \frac{1}{2} < Y < \frac{2}{3}\right] = \frac{769/13824}{79/432} = \mathbf{769/2528}$$

Q#6. The random variables X and Y are the jointly distributed as

$$f(x, y) = 24 x^2 y (1 - x) \quad 0 \leq x, y \leq 1$$

Obtain the marginal distribution of X and conditional distribution of Y .

Are X and Y independent?

Solution

$$f(x, y) = 24 x^2 y (1 - x) \quad 0 \leq x, y \leq 1$$

Marginal p. d. f of X

$$\begin{aligned}
f_1(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\
&= 24 \int_0^1 x^2 y (1 - x) dy \\
&= 24 x^2 \int_0^1 (y - xy) dy \\
&= 24 x^2 \left[\frac{y^2}{2} - x \frac{y^2}{2} \right]_0^1 \\
&= 24 x^2 \left[\frac{1}{2} - \frac{x}{2} \right] \\
&= 12 x^2 (1 - x) \quad 0 < x < 1
\end{aligned}$$

Marginal p. d. f of Y

$$\begin{aligned}
f_2(y) &= \int_{-\infty}^{\infty} f(x, y) dx \\
&= 24 y \int_0^1 x^2 (1 - x) dx
\end{aligned}$$

$$= 24 y \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1$$

$$= 24 y \left(\frac{1}{3} - \frac{1}{4} \right)$$

$$= 2 y$$

$$0 < y < 1$$

Conditional Distribution of Y

$$f(y|x) = \frac{f(x,y)}{f_1(x)}$$

$$= \frac{24x^2y(1-x)}{12x^2(1-x)} = 2y = f_2(y)$$

For independence

$$\begin{aligned} f_1(x) f_2(y) &= 12 x^2 (1-x) \times 2 y \\ &= 24 x^2 y (1-x) = f(x, y) \end{aligned}$$

Hence X and Y are independent.

Q#7. Let the Bivariate Continuous random variables (X, Y) have the joint p. d. f given by

$$\begin{aligned} f(x, y) &= x^2 + \frac{xy}{3}, & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ &= 0 & \text{elsewhere} \end{aligned}$$

Find Var(X), Var(Y) and Corr(X, Y)

Solution

Marginal p. d. f of X

$$\begin{aligned} f_1(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\ &= \int_0^2 \left(x^2 + \frac{xy}{3} \right) dy \\ &= \left[x^2 y + \frac{xy^2}{6} \right]_0^2 \\ &= 2x^2 + \frac{2x}{3} \end{aligned}$$

Marginal p. d. f of Y

$$\begin{aligned}
 f_2(y) &= \int_{-\infty}^{\infty} f(x, y) dx \\
 &= \int_0^1 \left(x^2 + \frac{xy}{3} \right) dx \\
 &= \left[\frac{x^3}{3} + \frac{y x^2}{6} \right]_0^1 \\
 &= \frac{1}{3} + \frac{1}{6} y
 \end{aligned}$$

$$\begin{aligned}
 E(X) &= \int_{-\infty}^{\infty} x f_1(x) dx \\
 &= \int_0^1 x \left(\frac{2}{3} x + 2x^2 \right) dx \\
 &= \int_0^1 \left(\frac{2}{3} x^2 + 2x^3 \right) dx \\
 &= \left[\frac{2x^3}{3 \cdot 3} + 2 \frac{x^4}{4} \right]_0^1 = 2/9 + 2/4 = 13/18
 \end{aligned}$$

$$\begin{aligned}
 E(X^2) &= \int_{-\infty}^{\infty} x^2 f_1(x) dx \\
 &= \int_0^1 x^2 \left(\frac{2}{3} x + 2x^2 \right) dx \\
 &= \int_0^1 \left(\frac{2}{3} x^3 + 2x^4 \right) dx \\
 &= \left[\frac{2x^4}{3 \cdot 4} + 2 \frac{x^5}{5} \right]_0^1 = 1/6 + 2/5 = 17/30
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(X) &= E(X^2) - (E(X))^2 \\
 &= 17/30 - \left(\frac{13}{18} \right)^2 = 73/1620
 \end{aligned}$$

$$E(Y) = \int_{-\infty}^{\infty} y f_2(y) dy$$

$$\begin{aligned}
 E(Y) &= \int_0^2 y \left(\frac{1}{3} + \frac{1}{6} y \right) dy \\
 &= \int_0^2 \left(\frac{y}{3} + \frac{y^2}{6} \right) dy \\
 &= \left[\frac{y^2}{6} + \frac{y^3}{18} \right]_0^2
 \end{aligned}$$

$$= \frac{4}{6} + \frac{8}{18} = \frac{10}{9}$$

$$\begin{aligned} E(Y^2) &= \int_0^2 y^2 \left(\frac{1}{3} + \frac{1}{6}y \right) dy \\ &= \int_0^2 \left(\frac{y^2}{3} + \frac{y^3}{6} \right) dy \\ &= \left[\frac{y^3}{9} + \frac{y^4}{24} \right]_0^2 \end{aligned}$$

$$= \frac{8}{9} + \frac{16}{24} = \frac{14}{9}$$

$$\begin{aligned} \text{Var}(Y) &= E(Y^2) - (E(Y))^2 \\ &= \frac{14}{9} - \left(\frac{10}{9} \right)^2 = \frac{26}{81} \end{aligned}$$

$$\begin{aligned} E(XY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy \\ &= \int_0^1 \int_0^2 x y \left(x^2 + \frac{xy}{3} \right) dx dy \\ &= \int_0^1 \int_0^2 \left(x^3 y + \frac{x^2 y^2}{3} \right) dx dy \\ &= \int_0^1 \left[x^3 \frac{y^2}{2} + x^2 \frac{y^3}{9} \right]_0^2 dx \\ &= \int_0^1 \left(2x^3 + \frac{8}{9}x^2 \right) dx \\ &= \left[\frac{2x^4}{4} + \frac{8}{9} \frac{x^3}{3} \right]_0^1 \\ &= \frac{1}{2} + \frac{8}{27} \\ &= \frac{43}{54} \end{aligned}$$

$$\begin{aligned} \text{Cov}(X, Y) &= E(XY) - E(X)E(Y) \\ &= \frac{43}{54} - \frac{13}{18} \times \frac{10}{9} = -\frac{1}{162} \end{aligned}$$

$$\text{Corr}(X, Y) = \rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

$$= \frac{-\frac{1}{162}}{\sqrt{(73/1620) \times (26/81)}} = -0.051$$

=====

Q# 8. Given the joint p. d. f, $f(x, y) = 2 - x - y$ $0 \leq x \leq 1, 0 \leq y \leq 1$
 $= 0$ elsewhere

Find the correlation coefficient between X and Y

Solution

Marginal p. d. f of X $f_1(x) = \int_{-\infty}^{\infty} f(x, y) dy$ $= \int_0^1 (2 - x - y) dy$ $= \left[2y - xy - \frac{y^2}{2} \right]_0^1$ $= 2 - x - \frac{1}{2}$ $= \frac{3}{2} - x \quad 0 < x < 1$	Marginal p. d. f of Y Similarly $f_2(y) = \frac{3}{2} - y \quad 0 < y < 1$
$E(X) = \int_{-\infty}^{\infty} x f_1(x) dx$ $= \int_0^1 x \left(\frac{3}{2} - x \right) dx$ $= \left[\frac{3x^2}{2 \cdot 2} - \frac{x^3}{3} \right]_0^1 = \frac{3}{4} - \frac{1}{3} = \frac{5}{12}$	Similarly $E(Y) = \frac{5}{12}$
$E(X^2) = \int_{-\infty}^{\infty} x^2 f_1(x) dx$ $= \int_0^1 x^2 \left(\frac{3}{2} - x \right) dx$ $= \int_0^1 \left(\frac{3}{2} x^2 - x^3 \right) dx$ $= \left[\frac{3x^3}{2 \cdot 3} - \frac{x^4}{4} \right]_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$	
$\text{Var}(X) = E(X^2) - (E(X))^2$ $= \frac{1}{4} - \left(\frac{5}{12} \right)^2 = \frac{11}{144}$	Similarly $\text{Var}(Y) = \frac{11}{144}$

$$\begin{aligned}
E(XY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y) dx dy \\
&= \int_0^1 \int_0^1 xy(2 - x - y) dx dy \\
&= \int_0^1 \int_0^1 (2xy - x^2y - xy^2) dx dy \\
&= \int_0^1 \left[2x \frac{y^2}{2} - x^2 \frac{y^2}{2} - x \frac{y^3}{3} \right]_0^1 dy \\
&= \int_0^1 \left(x - \frac{x^2}{2} - \frac{x}{3} \right) dx \\
&= \int_0^1 \left(\frac{2}{3}x - \frac{x^2}{2} \right) dx \\
&= \left[\frac{2}{3} \frac{x^2}{2} - \frac{x^3}{6} \right]_0^1 \\
&= \frac{1}{3} - \frac{1}{6} \\
&= \frac{1}{6}
\end{aligned}$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$= \frac{1}{6} - \frac{5}{12} \times \frac{5}{12} = -1/144$$

$$\text{Corr}(X, Y) = \rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

$$\rho = \frac{-1/144}{\sqrt{\left(\frac{11}{144}\right) \times \left(\frac{11}{144}\right)}} = -\frac{1}{144} \times \frac{144}{11} = -\frac{1}{11} = -0.091$$

=====

Q#9. Let X and Y are independent random variables with joint p.d.f's

$$f(x, y) = x+y \quad 0 \leq x \leq 1, 0 \leq y \leq 1$$

= 0

elsewhere

Find $E(XY)$ and $E(X+Y)$

(PU 2019)

Solution

Marginal p. d. f of X

$$\begin{aligned}
 f_1(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\
 &= \int_0^1 (x + y) dy \\
 &= \left[xy + \frac{y^2}{2} \right]_0^1 \\
 &= x + \frac{1}{2} \quad 0 < x < 1
 \end{aligned}$$

Marginal p. d. f of Y

$$\begin{aligned}
 f_2(y) &= \int_{-\infty}^{\infty} f(x, y) dx \\
 &= \int_0^1 (x + y) dx \\
 &= \left[\frac{x^2}{2} + xy \right]_0^1 \\
 &= y + \frac{1}{2} \quad 0 < y < 1
 \end{aligned}$$

Expectation of X and Y

$$\begin{aligned}
 E(X) &= \int_0^1 x f_1(x) dx \\
 &= \int_0^1 x \left(x + \frac{1}{2} \right) dx \\
 &= \int_0^1 \left(x^2 + \frac{1}{2}x \right) dx \\
 &= \left[\frac{x^3}{3} + \frac{1}{2} \frac{x^2}{2} \right]_0^1 \\
 &= \frac{1}{3} + \frac{1}{4} = \frac{7}{12}
 \end{aligned}$$

$$\begin{aligned}
 E(Y) &= \int_0^1 y f_2(y) dy \\
 &= \int_0^1 y \left(y + \frac{1}{2} \right) dy \\
 &= \int_0^1 \left(y^2 + \frac{1}{2}y \right) dy \\
 &= \left[\frac{y^3}{3} + \frac{1}{2} \frac{y^2}{2} \right]_0^1 \\
 &= \frac{1}{3} + \frac{1}{4} = \frac{7}{12}
 \end{aligned}$$

$$E(XY) = E(X) E(Y)$$

$$= \frac{7}{12} \times \frac{7}{12}$$

$$= 49/144 = 0.34$$

$$E(X+Y) = E(X) + E(Y)$$

$$= \frac{7}{12} + \frac{7}{12}$$

$$= 14/12 = 1.17$$

=====

Q# 10. Let X and Y be independent random variables with joint p.d.f's

$$f(x, y) = \frac{x(1+3y^2)}{4}$$

$$0 < x < 2, \quad 0 < y < 1$$

= 0

elsewhere

Find $E(X)$, $E(Y)$, $E(X+Y)$ $E(XY)$

Solution

Marginal p. d. f of X

$$\begin{aligned}
 f_1(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\
 &= \int_0^2 \frac{x(1+3y^2)}{4} dy \\
 &= \frac{1}{4} \left[xy + 3x \frac{y^3}{3} \right]_0^1 \\
 &= \frac{1}{4} (x + x) \\
 &= \frac{1}{2} x
 \end{aligned}$$

Marginal p. d. f of Y

$$\begin{aligned}
 f_2(y) &= \int_{-\infty}^{\infty} f(x, y) dx \\
 &= \int_0^2 \frac{x(1+3y^2)}{4} dx \\
 &= \frac{(1+3y^2)}{4} \left[\frac{x^2}{2} \right]_0^1 \\
 &= \frac{(1+3y^2)}{2}
 \end{aligned}$$

$$\begin{aligned}
 E(X) &= \int_0^2 x f_1(x) dx \\
 &= \int_0^2 x \cdot \frac{1}{2} x dx \\
 &= \frac{1}{2} \int_0^2 x^2 dx \\
 &= \frac{1}{2} \left[\frac{x^3}{3} \right]_0^2 \\
 &= \frac{4}{3}
 \end{aligned}$$

$$\begin{aligned}
 E(Y) &= \int_0^1 y f_2(y) dy \\
 &= \int_0^1 y \cdot \frac{(1+3y^2)}{2} dy \\
 &= \frac{1}{2} \int_0^1 (y + 3y^3) dy \\
 &= \frac{1}{2} \left[\frac{y^2}{2} + \frac{3y^4}{4} \right]_0^1 \\
 &= \frac{1}{2} \left(\frac{1}{2} + \frac{3}{4} \right) = \frac{5}{8}
 \end{aligned}$$

$$E(X+Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x+y) f(x, y) dx dy$$

$$\begin{aligned}
 E(X+Y) &= \int_0^2 \int_0^1 (x+y) \frac{x(1+3y^2)}{4} dx dy \\
 &= \int_0^2 \int_0^1 x \frac{x(1+3y^2)}{4} dx dy + \int_0^2 \int_0^1 y \frac{x(1+3y^2)}{4} dx dy \\
 &= \frac{1}{4} \int_0^2 \int_0^1 (x^2 + 3x^2 y^2) dx dy + \frac{1}{4} \int_0^2 \int_0^1 (xy + 3xy^3) dx dy \\
 &= \frac{1}{4} \int_0^2 \left[x^2 y + 3x^2 \frac{y^3}{3} \right]_0^1 dx + \frac{1}{4} \int_0^2 \left[x \frac{y^2}{2} + 3 \frac{xy^4}{4} \right]_0^1 dx
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4} \int_0^2 (2x^2) dx + \frac{1}{4} \int_0^2 \left(\frac{x}{2} + \frac{3x}{4} \right) dx \\
&= \frac{1}{2} \int_0^2 x^2 dx + \frac{1}{16} \int_0^2 (5x) dx \\
&= \frac{1}{2} \left[\frac{x^3}{3} \right]_0^2 + \frac{5}{16} \left[\frac{x^2}{2} \right]_0^2 \\
&= \frac{1}{2} \times \frac{8}{3} + \frac{5}{16} \times 2 = \frac{4}{3} + \frac{5}{8} = \frac{47}{24}
\end{aligned}$$

$$\begin{aligned}
E(XY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y) dx dy \\
&= \int_0^2 \int_0^1 xy \frac{x(1+3y^2)}{4} dx dy \\
&= \frac{1}{4} \int_0^2 \int_0^1 (x^2 y + 3x^2 y^3) dx dy \\
&= \frac{1}{4} \int_0^2 \left[x^2 \frac{y^2}{2} + 3x^2 \frac{y^4}{4} \right]_0^1 dx \\
&= \frac{1}{4} \int_0^2 \left(\frac{x^2}{2} + \frac{3}{4} x^2 \right) dx \\
&= \frac{1}{16} \int_0^2 (5x^2) dx \\
&= \frac{5}{16} \left[\frac{x^3}{3} \right]_0^2 = \frac{5}{16} \left(\frac{8}{3} \right) \\
&= \frac{5}{6}
\end{aligned}$$

Q#11. The joint p. d. of two dimensional random **variable X and Y** is given bellow

$$f(x, y) = \frac{1}{8} (6 - x - y)$$

$$0 \leq x \leq 2, 2 \leq y \leq 4$$

$$= 0$$

elsewhere

Calculate $VAR(X/Y)$

(PU 2013)

Solution

Marginal p. d. f of Y

$$\begin{aligned}
 f_2(y) &= \frac{1}{8} \int_2^4 (6 - x - y) dx & 0 < x < 2 \\
 &= \frac{1}{8} \left[6x - \frac{x^2}{2} - xy \right]_0^2 \\
 &= \frac{1}{8} [\{12 - 2 - 2y\} - 0] \\
 &= \frac{1}{8} [10 - 2y] & 0 < x < 2 \\
 &= \frac{1}{4} [5 - y] & 2 < y < 4 \\
 &= 0 & \text{elsewhere}
 \end{aligned}$$

The conditional p. d. f $f(x|y)$

$$\begin{aligned}
 f(x|y) &= \frac{f(x,y)}{f_2(y)} \\
 &= \frac{\frac{1}{8}(6-x-y)}{\frac{1}{4}[5-x]}
 \end{aligned}$$

$$f(x|y) = \frac{(6-x-y)}{2[5-x]}$$

$$\begin{aligned}
 E(X/Y) &= \int_{-\infty}^{\infty} x f(x|y) dx \\
 &= \frac{1}{2} \int_0^2 x \frac{(6-x-y)}{(5-y)} dx \\
 &= \frac{1}{2(5-y)} \int_0^2 (6x - x^2 - xy) dx \\
 &= \frac{1}{2(5-y)} \left[6 \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^2}{2} y \right]_0^2 \\
 &= \frac{1}{2(5-y)} \left[12 - \frac{8}{3} - 2y \right] \\
 &= \frac{(28-6y)}{6(5-y)}
 \end{aligned}$$

$$E(X/Y) = \frac{(14-3y)}{3(5-y)}$$

$$E(X^2|Y) = \int_{-\infty}^{\infty} x^2 f(x|y) dx$$

$$\begin{aligned}
&= \frac{1}{2} \int_0^2 x^2 \frac{(6-x-y)}{(5-y)} dx \\
&= \frac{1}{2(5-y)} \int_0^2 (6x^2 - x^3 - x^2 y) dx \\
&= \frac{1}{2(5-y)} \left[\frac{6x^3}{3} - \frac{x^4}{4} - \frac{x^3}{3} y \right]_0^2 \\
&= \frac{1}{2(5-y)} \left[16 - 4 - \frac{8}{3} y \right] \\
&= \frac{1}{2(5-y)} \frac{(36-8y)}{3} = \frac{4}{2(5-y)} \frac{(9-2y)}{3}
\end{aligned}$$

$$E(X^2|Y) = \frac{2}{3} \frac{(9-2y)}{(5-y)}$$

$$\text{Var}(X|Y) = E(X^2|Y) - [E(X|Y)]^2$$

$$\begin{aligned}
&= \frac{2}{3} \frac{(9-2y)}{(5-y)} - \left[\frac{(14-3y)}{3(5-y)} \right]^2 \\
&= \frac{1}{3(5-y)} \left[2(9-2y) - \frac{(14-3y)^2}{3(5-y)} \right] \\
&= \frac{1}{9(5-y)^2} [6(9-2y)(5-y) - (14-3y)^2] \\
&= \frac{1}{9(5-y)^2} [270 - 114y + 12y^2 - 196 - 9y^2 + 84y] \\
&= \frac{1}{9(5-y)^2} [3y^2 - 30y + 74]
\end{aligned}$$

=====

Bivariate Normal Distribution

The Bivariate Normal probability density function for the continuous random variable X and Y is defined as

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-m_1}{\sigma_1}\right)^2 + \left(\frac{y-m_2}{\sigma_2}\right)^2 - 2\rho\left(\frac{x-m_1}{\sigma_1}\right)\left(\frac{y-m_2}{\sigma_2}\right)\right\}\right]$$

Q# **Show that marginal density of 'X' in Bivariate Normal distribution follows univariate Normal distribution. (PU2016), (PU2017)**

Proof

The Bivariate Normal probability density function for the continuous random variable X and Y is defined as

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-m_1}{\sigma_1}\right)^2 + \left(\frac{y-m_2}{\sigma_2}\right)^2 - 2\rho\left(\frac{x-m_1}{\sigma_1}\right)\left(\frac{y-m_2}{\sigma_2}\right)\right\}\right]$$

The marginal distribution of random variable X is

$$f_1(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

Put $u = \frac{y-m_2}{\sigma_2}$ then $dy = \sigma_2 du$ therefore

$$\begin{aligned} f_1(x) &= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-m_1}{\sigma_1}\right)^2 + u^2 - 2\rho\left(\frac{x-m_1}{\sigma_1}\right)u\right\}\right] \sigma_2 du \\ &= \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-m_1}{\sigma_1}\right)^2 + u^2 - 2\rho\left(\frac{x-m_1}{\sigma_1}\right)u + \rho^2\left(\frac{x-m_1}{\sigma_1}\right)^2 - \rho^2\left(\frac{x-m_1}{\sigma_1}\right)^2\right\}\right] du \\ &= \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-m_1}{\sigma_1}\right)^2 (1-\rho^2) + \{u^2 - 2\rho\left(\frac{x-m_1}{\sigma_1}\right)u + \rho^2\left(\frac{x-m_1}{\sigma_1}\right)^2\}\right\}\right] du \\ &= \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\left\{\left(\frac{x-m_1}{\sigma_1}\right)^2 - \frac{1}{2(1-\rho^2)}\{u^2 - 2\rho\left(\frac{x-m_1}{\sigma_1}\right)u + \rho^2\left(\frac{x-m_1}{\sigma_1}\right)^2\}\right\}\right] du \end{aligned}$$

$$= \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2} \left\{ \left(\frac{x-m_1}{\sigma_1} \right)^2 - \frac{1}{2(1-\rho^2)} \left\{ u - \rho \left(\frac{x-m_1}{\sigma_1} \right) \right\}^2 \right\} \right] du$$

$$= \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2} \left(\frac{x-m_1}{\sigma_1} \right)^2 \right] \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2(1-\rho^2)} \left\{ u - \rho \left(\frac{x-m_1}{\sigma_1} \right) \right\}^2 \right] du$$

Put $\frac{1}{\sqrt{1-\rho^2}} \left\{ u - \rho \left(\frac{x-m_1}{\sigma_1} \right) \right\} = t$ then $du = \sqrt{1-\rho^2} dt$

$$f_1(x) = \frac{1}{2\pi\sigma_1\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2} \left(\frac{x-m_1}{\sigma_1} \right)^2 \right] \int_{-\infty}^{\infty} \exp \left(-\frac{t^2}{2} \right) \sqrt{1-\rho^2} dt$$

$$f_1(x) = \frac{1}{2\pi\sigma_1} \exp \left[-\frac{1}{2} \left(\frac{x-m_1}{\sigma_1} \right)^2 \right] \sqrt{2\pi} \quad \text{since } \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} dz = \sqrt{2\pi}$$

$$= \frac{1}{\sigma_1\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x-m_1}{\sigma_1} \right)^2 \right]$$

Hence $X \sim N(m_1, \sigma_1^2)$

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Prove that total area is one in Bivariate Normal Distribution.

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left[-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{x-m_1}{\sigma_1} \right)^2 + \left(\frac{y-m_2}{\sigma_2} \right)^2 - 2\rho \left(\frac{x-m_1}{\sigma_1} \right) \left(\frac{y-m_2}{\sigma_2} \right) \right\} \right]$$

Put $Z = \frac{x-m_1}{\sigma_1}$ $W = \frac{y-m_2}{\sigma_2}$

The Jacobian transformation is given as

$$J = \frac{\partial(x, y)}{\partial(z, w)} = \begin{vmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial w} \end{vmatrix}$$

$$\frac{\partial x}{\partial z} = \sigma_1, \quad \frac{\partial x}{\partial w} = 0$$

$$\frac{\partial y}{\partial z} = 0, \quad \frac{\partial y}{\partial w} = \sigma_2$$

$$J = \begin{vmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{vmatrix} = \sigma_1 \sigma_2 \quad dx dy = \sigma_1 \sigma_2 dz dw$$

Now total area = $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy$

$$\begin{aligned}
 &= \frac{\sigma_1 \sigma_2}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2(1-\rho^2)} \{ (z)^2 + (w)^2 - 2\rho(z)(w) \} \right] dz dw \\
 &= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2(1-\rho^2)} \{ (z)^2 - 2\rho(z)(w) + \rho^2 w^2 - \rho^2 w^2 + (w)^2 \} \right] dz dw \\
 &= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2(1-\rho^2)} \{ (z - \rho w)^2 + w^2(1 - \rho^2) \} \right] dz dw \\
 &= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2(1-\rho^2)} (z - \rho w)^2 \right] \exp \left\{ -\frac{1}{2(1-\rho^2)} w^2(1 - \rho^2) \right\} dz dw \\
 &= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2} \left(\frac{z - \rho w}{\sqrt{1-\rho^2}} \right)^2 \right] \exp \left[-\frac{w^2}{2} \right] dz dw
 \end{aligned}$$

Put $\frac{z - \rho w}{\sqrt{1-\rho^2}} = v$ and $w = t$

The Jacobian transformation is given as

$$J = \frac{\partial(z, w)}{\partial(v, t)} = \begin{vmatrix} \frac{\partial z}{\partial v} & \frac{\partial z}{\partial t} \\ \frac{\partial w}{\partial v} & \frac{\partial w}{\partial t} \end{vmatrix}$$

$$\frac{\partial z}{\partial v} = \sqrt{1-\rho^2} \quad \frac{\partial z}{\partial t} = 0$$

$$\frac{\partial w}{\partial v} = -\frac{\sqrt{1-\rho^2}}{\rho} \quad \frac{\partial w}{\partial t} = 1$$

$$J = \begin{vmatrix} \sqrt{1-\rho^2} & 0 \\ -\frac{\sqrt{1-\rho^2}}{\rho} & 1 \end{vmatrix} = \sqrt{1-\rho^2}$$

$$|J| dz dw = \sqrt{1-\rho^2} dv dt$$

Therefore

$$\begin{aligned}
 \text{Total Area} &= \frac{\sqrt{1-\rho^2}}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp \left[-\frac{1}{2} (v)^2 \right] \exp \left[-\frac{t^2}{2} \right] dv dt \\
 &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp \left(-\frac{v^2}{2} \right) dv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp \left(-\frac{t^2}{2} \right) dt
 \end{aligned}$$

$$= 1 \times 1$$

$$= 1$$

Hence the proof.

Moment Generating Function of Bivariate Normal Distribution

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left\{\left(\frac{x-m_1}{\sigma_1}\right)^2 + \left(\frac{y-m_2}{\sigma_2}\right)^2 - 2\rho\left(\frac{x-m_1}{\sigma_1}\right)\left(\frac{y-m_2}{\sigma_2}\right)\right\}\right]$$

$$\text{Put } Z = \frac{x-m_1}{\sigma_1} \quad W = \frac{y-m_2}{\sigma_2}$$

We get

$$f(z, w) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\{z^2 + w^2 - 2\rho zw\}\right]$$

$$M_0(t_1, t_2) = E(e^{t_1 z + t_2 w})$$

$$\begin{aligned} &= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{t_1 z + t_2 w} e^{-\frac{1}{2(1-\rho^2)}\{z^2 + w^2 - 2\rho zw\}} dz dw \\ &= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{\{t_1 z + t_2 w - \frac{1}{2(1-\rho^2)}\{z^2 + w^2 - 2\rho zw\}\}} dz dw \end{aligned}$$

Consider the exponent

$$\begin{aligned} &t_1 z + t_2 w - \frac{1}{2(1-\rho^2)}\{z^2 + w^2 - 2\rho zw\} \\ &= -\frac{1}{2(1-\rho^2)}[z^2 + w^2 - 2\rho zw - 2(1-\rho^2)(t_1 z + t_2 w)] \\ &= -\frac{1}{2(1-\rho^2)}[(z - t_1 - \rho t_2)^2 + (w - t_2 - \rho t_1)^2 - 2\rho(z - t_1 - \rho t_2)(w - t_2 - \rho t_1) - (1-\rho^2)(t_1^2 + t_2^2 + 2\rho t_1 t_2)] \end{aligned}$$

Now using transformation $U = z - t_1 - \rho t_2$, $V = w - t_2 - \rho t_1$

$$J = \frac{\partial(z, w)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \\ \frac{\partial w}{\partial u} & \frac{\partial w}{\partial v} \end{vmatrix} \quad \frac{\partial z}{\partial u} = 1 \quad \frac{\partial z}{\partial v} = 0 \quad \frac{\partial w}{\partial u} = 0 \quad \frac{\partial w}{\partial v} = 1$$

$$J = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

So index of exponent is

$$= -\frac{1}{2(1-\rho^2)} [u^2 + v^2 - 2\rho uv - (1-\rho^2)(t_1^2 + t_2^2 + 2\rho t_1 t_2)]$$

$$\begin{aligned} M_0(t_1, t_2) &= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{2(1-\rho^2)}[u^2+v^2-2\rho uv-(1-\rho^2)(t_1^2+t_2^2+2\rho t_1 t_2)]} du dv \\ &= e^{\frac{1}{2}(t_1^2+t_2^2+2\rho t_1 t_2)} \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{2(1-\rho^2)}[u^2+v^2-2\rho uv]} du dv \end{aligned}$$

$$M_0(t_1, t_2) = e^{\frac{1}{2}(t_1^2+t_2^2+2\rho t_1 t_2)}$$

$$z = \frac{x-m_1}{\sigma_1} \quad x = m_1 + \sigma_1 z$$

$$w = \frac{y-m_2}{\sigma_2} \quad y = m_2 + \sigma_2 w$$

Hence m. f of X and Y about origin

$$M_0(t_1, t_2) = e^{m_1 t_1 + m_2 t_2 + \frac{1}{2}(t_1^2 \sigma_1^2 + t_2^2 \sigma_2^2 + 2\rho t_1 t_2 \sigma_1 \sigma_2)}$$

The cumulant generating function of X and Y

$$K(t_1, t_2) = \log M_0(t_1, t_2)$$

$$= m_1 t_1 + m_2 t_2 + \frac{1}{2}(t_1^2 \sigma_1^2 + t_2^2 \sigma_2^2 + 2\rho t_1 t_2 \sigma_1 \sigma_2)$$

$$K_{10} = m_1 \quad K_{01} = m_2$$

$$K_{20} = \sigma_1^2 \quad K_{02} = \sigma_2^2 \quad K_{11} = \rho \sigma_1 \sigma_2$$

$$\text{Corr}(X, Y) = \frac{K_{11}}{\sqrt{K_{20} K_{02}}} = \frac{\rho \sigma_1 \sigma_2}{\sigma_1 \sigma_2} = \rho$$

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